

pgAdmin 4

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - Login/Group Roles
 - Tablespaces

Dependencies X Dependents X Processes X dvdrental /postgres... X dvdrental /postgres@PostgreSQL 18* X

dvdrental /postgres@PostgreSQL 18

Query Query History

```
192
193
194 -- lab 1: blocking bằng row-level lock
195
196
197 -- session a - bước 1
198 begin;
199 update film set rental_rate = 4.99 where film_id = 1;
200
201 -- session b - bước 2, câu update sẽ bị chặn
202 begin;
203 update film set rental_rate = 5.99 where film_id = 1;
204
205 -- session a - bước 3
206 commit;
207
208 -- session b - bước 4, chạy sau khi update tự hoàn tất
209 commit;
210
211
212 -- lab 2: non-repeatable read
213
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
WARNING: there is no transaction in progress
COMMIT

Query returned successfully in 133 msec.

Total rows: Query complete 00:00:00.133 LF Ln 194, Col 1

pgAdmin 4

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - Login/Group Roles
 - Tablespaces

Dependencies X Dependents X Processes X dvdrental /postgres... X dvdrental /postgres@PostgreSQL 18* X

dvdrental /postgres@PostgreSQL 18

Query Query History

```
210
211
212 -- lab 2: non-repeatable read
213
214
215 -- session a - bước 1
216 begin isolation level read committed;
217 select rental_rate from film where film_id = 10;
218
219 -- session b - bước 2
220 begin;
221 update film set rental_rate = 0.99 where film_id = 10;
222 commit;
223
224 -- session a - bước 3 và 4
225 select rental_rate from film where film_id = 10;
226 commit;
227
228
229 -- lab 3: tạo deadlock
230
231
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
WARNING: there is no transaction in progress
COMMIT

Query returned successfully in 40 msec.

Total rows: Query complete 00:00:00.040 LF Ln 212, Col 1

pgAdmin 4

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - Login/Group Roles
 - Tablespaces

Dependencies X Dependents X Processes X dvdrental / postgre... X dvdrental / postgres@PostgreSQL 18* X

Query

```
-- lab 3: tạo deadlock
-- =====
-- session a - bước 1
begin;
update film set rental_rate = 2.99 where film_id = 100;

-- session b - bước 2
begin;
update film set rental_rate = 2.99 where film_id = 200;

-- session a - bước 3, câu update sẽ chờ
update film set rental_rate = 3.99 where film_id = 200;

-- session b - bước 4, postgresql trả lỗi deadlock detected
update film set rental_rate = 3.99 where film_id = 100;

-- session b - bước 5
rollback;

-- session a - bước 5
commit;
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
WARNING: there is no transaction in progress
COMMIT

Query returned successfully in 26 msec.

Total rows: Query complete 00:00:00.026 LF Ln 229, Col 1

pgAdmin 4

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Castes
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - Login/Group Roles
 - Tablespaces

Dependencies X Dependents X Processes X dvdrental / postgre... X dvdrental / postgres@PostgreSQL 18* X

Query

```
set local session_replication_role = 'replica';
select * from inventory where inventory_id = 10 for update;
insert into rental (rental_date, inventory_id, customer_id, staff_id)
values (current_timestamp, 10, 1, 1)
returning rental_id;

-- session b - bước 2, câu select sẽ chờ
begin;
select * from inventory where inventory_id = 10 for update;
select exists (
    select 1
    from rental
    where inventory_id = 10
    and return_date is null
) as already_rented;

-- session a - bước 3
commit;

-- session b - bước 4, already_rented = true nên không insert
rollback;

-- sau cùng, chạy file restore.sql
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
WARNING: there is no transaction in progress
ROLLBACK

Query returned successfully in 38 msec.

Total rows: Query complete 00:00:00.038 LF Ln 280, Col 35